PHYS 104 – PHYSICS II

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FALL 2026

Lecture 1

Textbook:

Physics for Scientists and Engineers with Modern Physics
Raymond A. Serway and John W. Jewett, Jr.

Reference:

Sears and Zemansky's University Physics: With Modern Physics Textbook by Albert Lewis Ford, Hugh D. Young, and Roger A. Freedman

Grading System:

Assignments 10%

Quizzes (2 or 3) 7%

Midterm exam (One) 10%

Laboratory 33%

Final exam 40%



Chapter 35

The Nature of Light and the Principles of Ray Optics

The Nature of Light

- Before the beginning of the nineteenth century, light was considered to be a stream of particles.
- The particles were emitted by the object being viewed,
  - Newton was the chief architect of the particle theory of light.
  - He believed the particles left the object and stimulated the sense of sight upon entering the eyes.
- Christian Huygens argued the light might be some sort of a wave motion.
- Thomas Young (1801) provided the first clear demonstration of the wave nature of light
  - He showed that light rays interfere with each other
  - Such behavior could not be explained by particles

The Nature of Light

- During the nineteenth century, other developments led to the general acceptance of the wave theory of light
- Maxwell asserted that light was a form of high-frequency electromagnetic wave.
- Hertz confirmed Maxwell's predictions.
- Some experiments could not be explained by the wave nature of light.
- The photoelectric effect was a major phenomenon not explained by waves,
  - When light strikes a metal surface, electrons are sometimes ejected from the surface
  - The kinetic energy of the ejected electron is independent of the intensity of the light

Dual Nature of Light

- Einstein proposed an explanation of the photoelectric effect in 1905 using a model based on the concept of quantization developed by Max Planck (1858–1947) in 1900.
- The quantization model assumes the energy of a light wave is present in particles called photons; hence, the energy is said to be quantized.
- According to Einstein's theory, the energy of a photon is proportional to the frequency f of the electromagnetic wave:

$$\boldsymbol{E = hf}$$

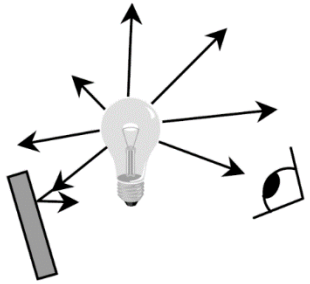
Where $h = 6.63 \times 10^{-34}\ \mathrm{J \cdot sec}$ is called Planck's constant.

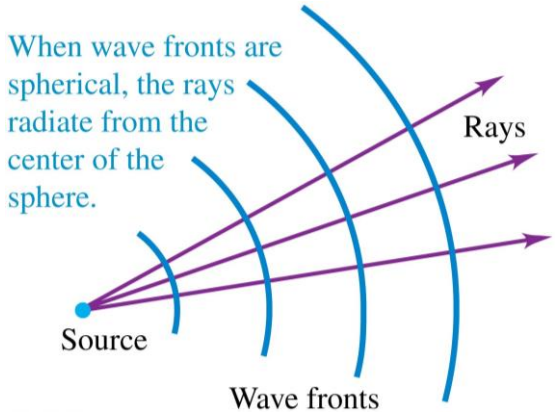
Sometimes light acts like a wave, and other times it acts like a particle.

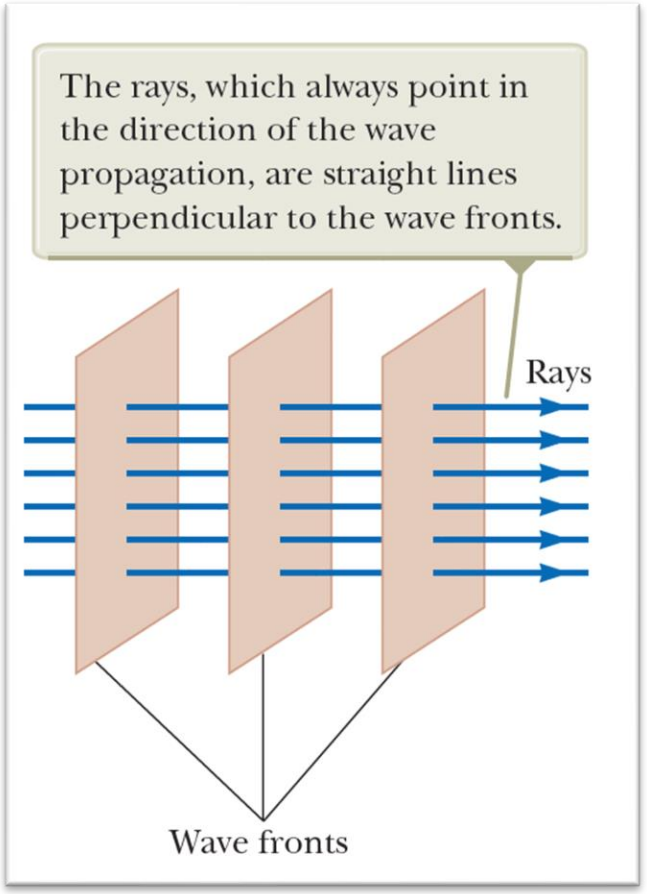
The Ray Approximation in Ray "Geometric" Optics

- Ray optics assumes light travels in a fixed direction in a straight line as it passes through a uniform medium and changes its direction when it meets the surface of a different medium or if the optical properties of the medium are nonuniform in either space or time.









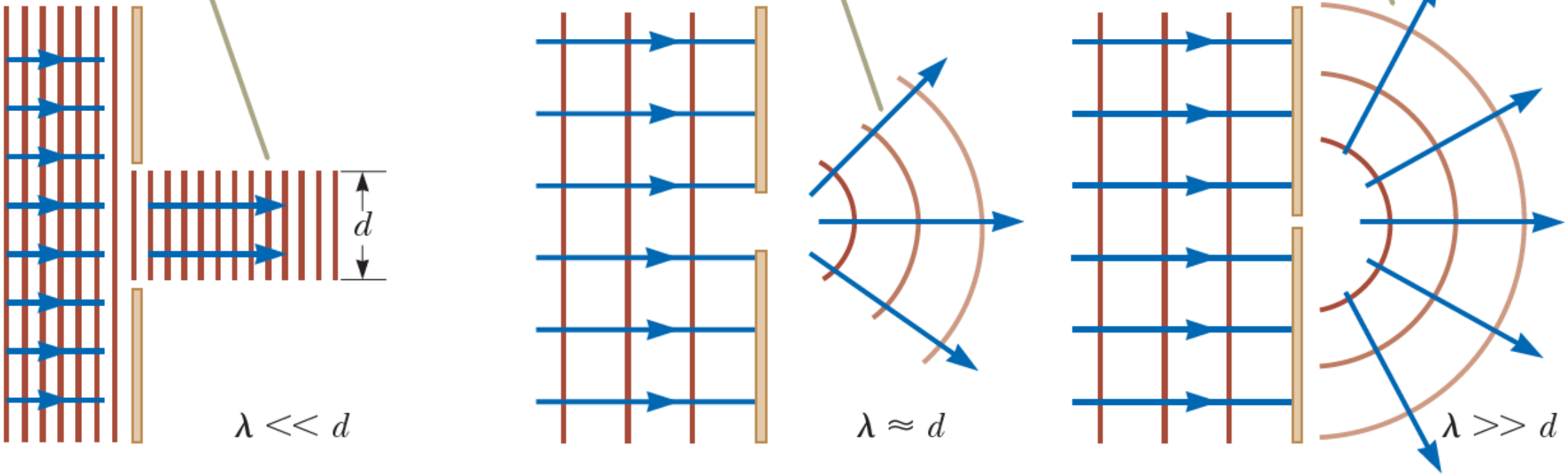
- To understand this approximation, first notice that the rays of a given wave are straight lines perpendicular to the wave fronts.
- We assume that a wave moving through a medium travels in a straight line in the direction of its rays.

Ray Approximation at a Barrier

When $\lambda << d$, the rays continue in a straight-line path and the ray approximation remains valid.

When $\lambda \approx d$, the rays spread out after passing through the opening.

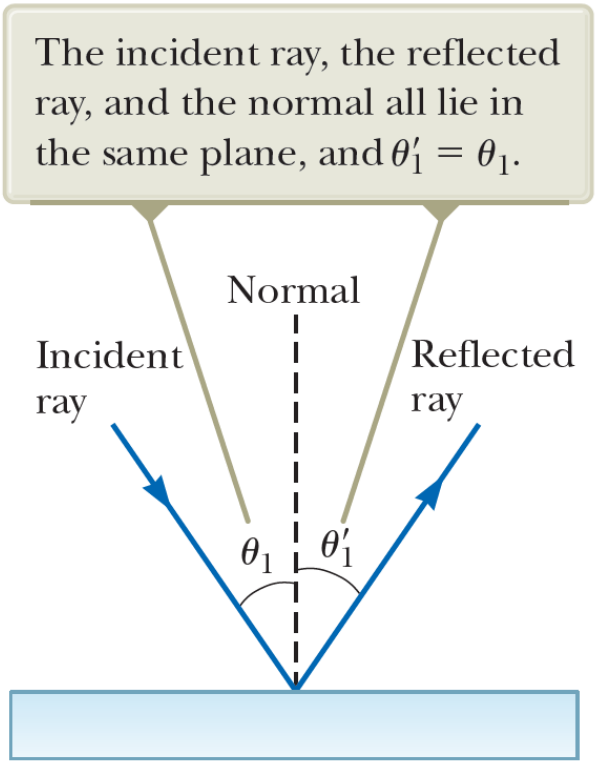
When $\lambda >> d$, the opening behaves as a point source emitting spherical waves.



Diffraction

Reflection of Light

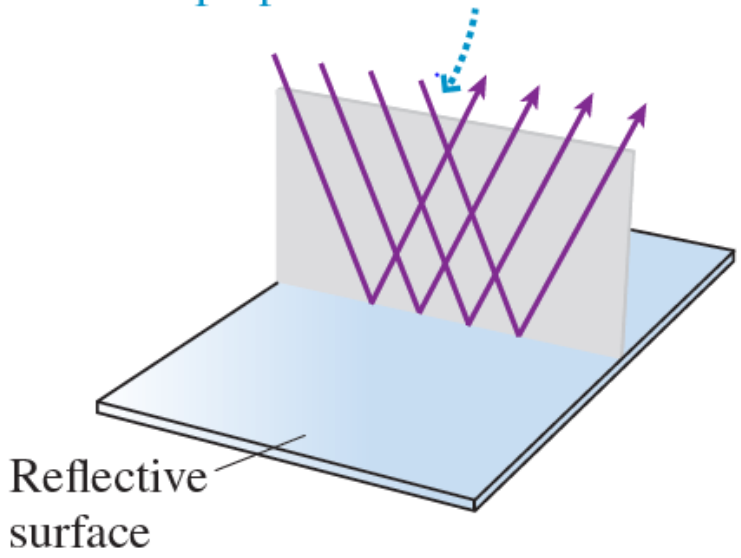
- It is the rebounding back of light when it meets a reflecting surface.
- Angle of incidence: angle between incident ray and normal to surface.
- Angle of reflection: angle between reflected ray and normal to surface
- Reflection rules (laws):

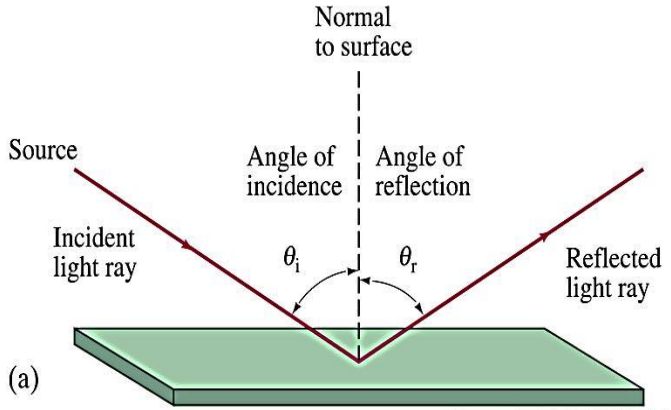


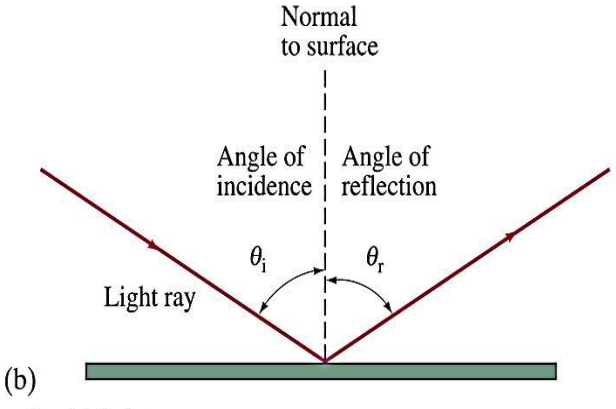
Rule I: $\boldsymbol{Angle\ of\ incidence\ (\theta_1) = (\theta_1')\ Angle\ of\ reflection}$

Rule II: $\boldsymbol{The\ incident\ light\ ray, reflected\ light\ ray\ and\ the\ normal\ to\ reflecting\ surface\ must\ all\ lie\ in\ the\ same\ plane}$

Both the incident and reflected rays lie in a plane that is perpendicular to the surface.







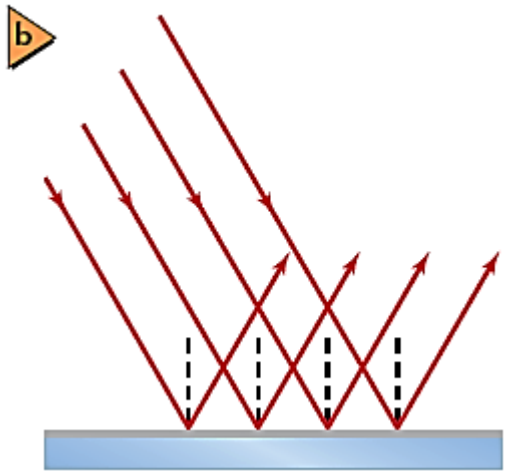
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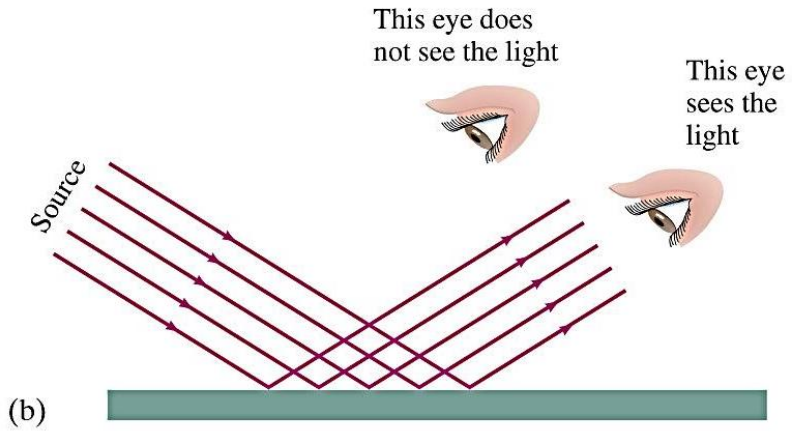
- If the incident ray is normal to the surface ($\theta_1 = 0$), it will be reflected on itself ($\theta_1' = \theta_1 = 0$)
- The path of a light ray is reversible.

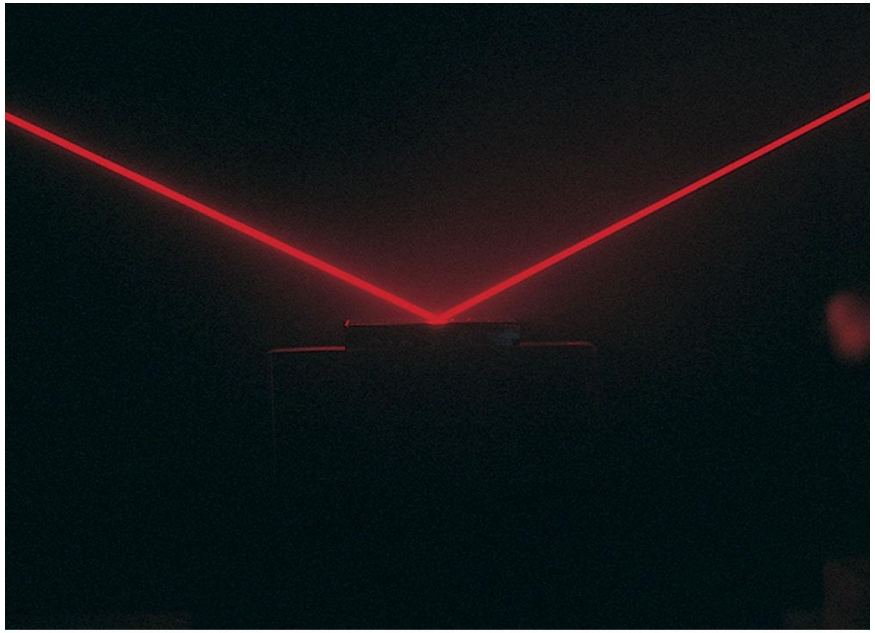
Types of Reflection

1- Specular "Regular" Reflection

- Reflection on a smooth surface.
- Reflected rays are parallel to each other.





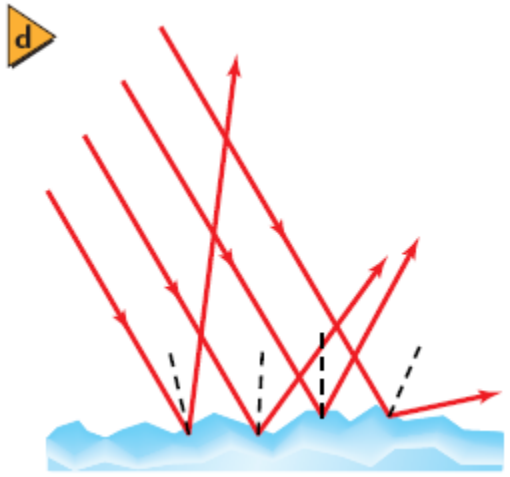


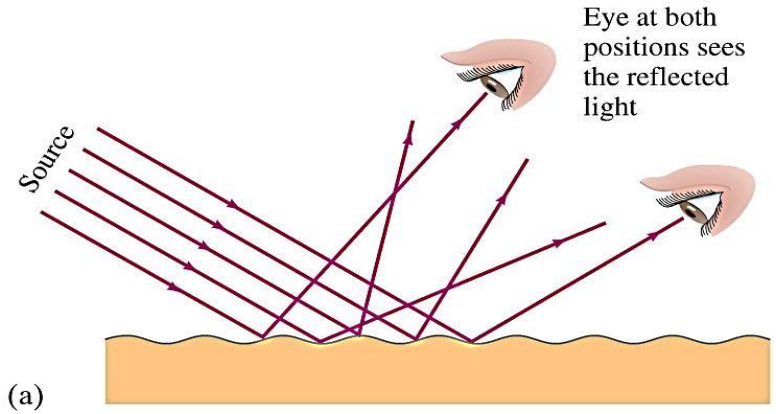
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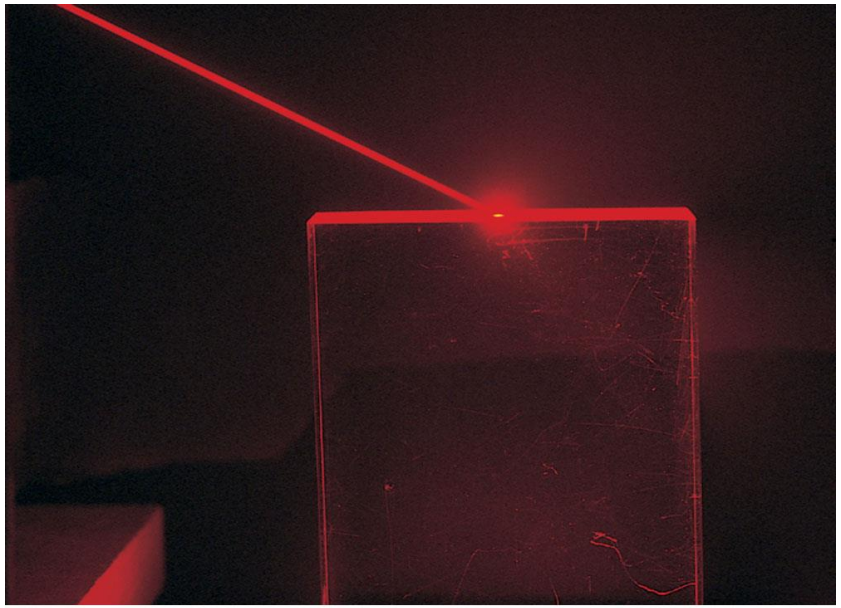
Types of Reflection

2- Diffuse Reflection

- Reflection on a rough surface.
- Reflected rays are in random directions.





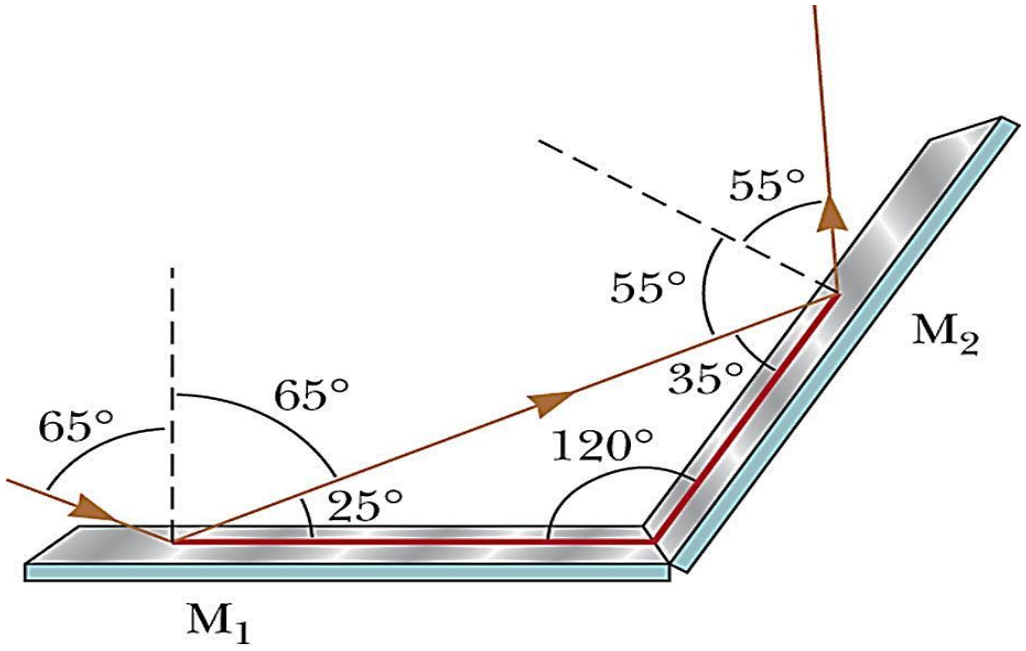


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Example 1 (Multiple reflections):

Two mirrors make an angle of 120° with each other. A ray is incident on mirror M_1 at an angle of 65° to the normal. Find the angle the ray makes with the normal to M_2 after it is reflected from both mirrors.

Solution



Retroreflection

- If the angle between two mirrors is 90°, the reflected beam returns to the source parallel to its original path.

